

Universal Plastic Shredder V2.0 Electrical System Test Plan

Course: ECE 412 Senior Project Development, Portland State University

Sponsor: MME / Ichor Systems (Bridget Lannigan)

Team: Bao Nguyen, Fearghus Tyler, Yaqoub Rabiah, Fox Kang

Purpose

Verify the electrical control system of the Universal Plastic Shredder V2.0 using top-down and bottom-up testing methods.

Top-Down Test Summary

Requirement	Pass	Fail	Notes
M1 — Shreds to ~1"	■	■	
M2 — VFD controllable	■	■	
M4 — E-stop shutdown	■	■	
M6 — Lid interlock	■	■	
M7 — 70–90 RPM	■	■	
M9 — HMI display	■	■	
S1 — Status lights	■	■	

BU-01: AC Power Input and Fuse Distribution

#	Input	Expected Output	Pass	Fail	N/A	Comments
1	Main ON	Voltage at fuse input	■	■	■	
2	Main ON	Voltage at PSU input	■	■	■	
3	Main ON	Voltage at PLC input	■	■	■	
4	Remove fuse	PSU off, PLC unaffected	■	■	■	

BU-05: Two-Hand Start Safety

#	Input	Expected Output	Pass	Fail	N/A	Comments
1	Forward only	No start	■	■	■	
2	SW1 only	No start	■	■	■	
3	SW2 only	No start	■	■	■	
4	SW1+SW2+Fwd	Motor starts	■	■	■	

Requirements Coverage Summary

Req ID	Requirement	Result
M1	Shred to ~1"	■ Pass ■ Fail
M4	E-stop safe shutdown	■ Pass ■ Fail
M6	Safety interlock	■ Pass ■ Fail
M7	RPM range	■ Pass ■ Fail
M9	HMI control	■ Pass ■ Fail
S1	Indicator lights	■ Pass ■ Fail